In [ ]:

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PAWAR SAKSHI MAHENDRA
ROLL NO - 49
PRACTICAL NO - 13
```

In [1]:

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
from tensorflow.keras.optimizers import Adam

# Load and preprocess the MNIST dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

X_train = X_train / 255.0
X_test = X_test / 255.0

# Define the model (FIXED ERROR)
model = Sequential([
    Flatten(input_shape=(28, 28)),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

# Compile the model
model.compile(optimizer=Adam(learning_rate=0.001),
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train the model
model.fit(X_train, y_train, batch_size=64, epochs=10, verbose=1)

# Evaluate the model (FIXED ERROR)
loss, accuracy = model.evaluate(X_test, y_test)

print(f"Test Loss: {loss}")
print(f"Test Accuracy: {accuracy}")
```

```
C:\Users\saksh\anaconda3\Lib\site-packages\keras\src\layers\reshaping\flatten.p
y:37: UserWarning: Do not pass an `input_shape`/`input_dim` argument to a laye
r. When using Sequential models, prefer using an `Input(shape)` object as the f
irst layer in the model instead.
  super().__init__(**kwargs)
```

```
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows fo
r TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. P
lease use WSL2 or the TensorFlow-DirectML plugin.
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 21s 14ms/step - accuracy: 0.9152 - loss: 0.3000
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 13s 14ms/step - accuracy: 0.9607 - loss: 0.1359
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 20s 14ms/step - accuracy: 0.9721 - loss: 0.0947
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 18s 11ms/step - accuracy: 0.9793 - loss: 0.0717
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 11s 12ms/step - accuracy: 0.9832 - loss: 0.0562
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 13s 14ms/step - accuracy: 0.9862 - loss: 0.0456
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 21s 14ms/step - accuracy: 0.9888 - loss: 0.0366
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 21s 14ms/step - accuracy: 0.9910 - loss: 0.0306
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 21s 14ms/step - accuracy: 0.9930 - loss: 0.0243
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 12s 13ms/step - accuracy: 0.9940 - loss: 0.0201
313/313 ━━━━━━━━━━━━━━━━━━━━ 5s 11ms/step - accuracy: 0.9777 - loss: 0.0702
Test Loss: 0.07015115022659302
Test Accuracy: 0.9776999950408936
```

In [ ]: